

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Backend Engineer with 5+ years building high-throughput **Node.js** and **TypeScript** systems. Led zero-downtime cloud migrations across **AWS** and **GCP**, saving \$60K annually in infrastructure costs. Deep expertise in **distributed systems**, **MongoDB**, and **serverless architectures**. Proven ability to own services end-to-end from design to operation, with strong focus on **observability**, **testing**, and **performance**. Experienced in translating **business requirements** into **scalable solutions** for fintech and enterprise applications.

CORE COMPETENCIES

- Node.js / TypeScript
- AWS Serverless (Lambda, SQS)
- Distributed Systems
- MongoDB / Firestore
- High-Quality API Design
- Cloud Migrations
- Observability & Tracing
- Build-and-Run Ownership

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Architected and executed **zero-downtime migration** of 20+ production applications from Parse/MongoDB to Firebase/GCP, demonstrating **build-and-run ownership** and reporting directly to CTO
- Designed real-time two-way sync between **MongoDB** and Firestore using **Node.js** and Google Pub/Sub for **distributed** message processing with full **observability**
- Built 3D visualization feature using **Node.js** and ffmpeg, directly increasing buyer conversion through improved customer experience
- Delivered **\$5,000/month savings** by migrating all services off **AWS** while maintaining **performance** and reliability
- Implemented Express Shipping feature end-to-end, generating additional \$40 revenue per order

MTailor

Mar 2022 – Jul 2022

Contractor · US Remote

- Built migration script processing complex data transformations from **MongoDB** to Firestore with comprehensive **testing** and validation
- Wrote technical documentation enabling cross-functional teams to work with **Node.js** and Python Firebase SDKs
- Trained Ops team on Firebase Dashboard, ensuring full **build-and-run** capability transfer

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Built real-time stock market pricing system querying **2M+ records** from Firestore with optimized batch operations and **performance** tuning
- Extended backend **APIs** using Firebase Cloud Functions and Express.js, focusing on **scalability** and reliability
- Delivered responsive PWA components using React and Tailwind, translating **business requirements** into user-facing features

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using Web3 and **Node.js**, handling complex **distributed** transaction flows
- Delivered React UIs from Figma designs under tight timelines, reporting directly to CTO

CodeBits

Jan 2020 – Jul 2021

Software Engineer · Uganda Remote

- Led team of 4 developers building legal tech systems using **distributed systems** architecture with Apache Kafka, Docker, and Kubernetes
- Architected microservices backend with event-driven workflows using Kafka and gRPC for **high-quality API** communication
- Built cross-platform mobile app using React Native with Firebase push notifications, ensuring **observability** across platforms
- Developed USSD service communicating with **Node.js** backend via gRPC, demonstrating **scalable solution** design

EDUCATION

Bachelor of Science in Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Languages: Node.js, TypeScript, Python, Go, Rust **Cloud:** AWS (Lambda, SQS, S3, EC2), GCP (Cloud Functions, Pub/Sub), Firebase
Databases: MongoDB, Firestore, SQL **Infrastructure:** Docker, Kubernetes, Terraform
Distributed Systems: Apache Kafka, gRPC, NATS Streaming, Event-driven Architecture **Frontend:** React, React Native, HTML, CSS, Tailwind